



PERSONAL RESEARCH BRIEF

# The Autonomic Nervous System Is Trainable

A personal note and some reading from Randy Graybeal, MSN, CRNA

Don,

Jason shared your note with me. I'm glad he did — it really resonated with me, both personally and clinically.

First, I just want to say — the way you described what you've been experiencing is remarkable. Not just the detail, but the quality of it. The crowd trigger for vasovagal episodes. The heart rate swings during hockey. Recognizing that lowering heart rate isn't the point when the underlying issue is a parasympathetic surge. Figuring out on your own that breathing prevents the cascade, and that Valsalva during exertion is a risk factor.

That kind of observation — understanding what your body is actually doing, not just what it feels like — is genuinely rare. Most people go through years of specialist visits before they can describe their own physiology that clearly. Some never get there. You're thinking about this the way I would as a clinician: what are the inputs, what are the outputs, and where does the system need support.

My 15-year-old went through about a year of autonomic dysfunction during one of the most challenging periods of their life. The symptoms were disruptive. The workups were inconclusive. The standard medical responses weren't connecting the dots.

We ended up on a path that was deceptively simple: increased salt intake to expand blood volume, a structured physical therapy program to recondition the autonomic response, and — honestly — patience while their nervous system recalibrated during a time when everything else in their world felt unstable.

**They're completely through it now. No residual symptoms. No medication.**

Our situations are different in a lot of ways — a teenager's autonomic nervous system is still developing, and the recovery timeline looks different than it does for an adult. But the underlying principle is something I keep seeing, both in the literature and in my own clinical experience:

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*The autonomic nervous system responds to **structured input**.  
And the people who tend to do well are the ones  
who are paying close attention to what their body is telling them.*

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What struck me is that you're not just managing this — you're learning from it. You mapped your own physiology, found an intervention that works, and tested it against your own experience. There's a growing body of research in interoception — the science of how accurately we sense what's happening inside our own bodies — and one of the consistent findings is that people who develop that kind of awareness tend to have better outcomes.

Not because awareness is magic, but because it lets you respond to what's actually happening instead of guessing. You're already doing that.

I spend a lot of time in this literature, both as a nurse anesthetist and because of what our family went through. Here's what I think you'd find interesting about what the current science shows.

## CURRENT EVIDENCE

### MECHANISM

## The Respiratory Component of the Vasovagal Cascade

Mennitto et al. · Psychosomatic Medicine · 2020 · RCT, n=160

This one caught my attention because of what you described. It turns out that even subtle hyperventilation — without any change in breathing rate — drives CO<sub>2</sub> depletion that contributes to the vasovagal cascade. Controlled breathing stabilizes end-tidal CO<sub>2</sub> and addresses the *respiratory* component of presyncope, which most standard workups don't look at. Changes in end-tidal CO<sub>2</sub> predicted vasovagal reactions (**OR = 0.57, p = .001**).

The interesting part: it was the *depth* of breathing that mattered, not the speed. Which lines up with what you've been doing.

## Five Weeks of Breathing Training Improves Baroreflex Sensitivity

Bellocchi et al. · European Journal of Medical Research · 2026 · Sham-controlled RCT

This just came out last month, and it's probably the most directly relevant paper for your situation. Five weeks of inspiratory muscle training produced **+4.7 ms/mmHg baroreflex sensitivity** ( $p=0.01$ ), **+20.5 n.u. vagal HRV** ( $p=0.02$ ), along with improvements in expiratory time, anti-inflammatory markers (IL-10), and sleep quality. The baroreflex is essentially the feedback loop that catches blood pressure before it drops too far during a vagal surge — which is exactly what's happening in vasovagal episodes. What's encouraging about this study is the timeline. Five weeks of training, measurable improvement in the specific system that matters.

### FRAMEWORK SHIFT

## From Relaxation to Regulation

Giorgi & Tedeschi · Acta Neurologica Belgica · 2025 · Scoping Review

This is the paper that shaped a lot of how I think about this. The field has been shifting away from "how do we calm people down" toward something more nuanced: *autonomic flexibility* — the ability of the nervous system to shift smoothly between sympathetic and parasympathetic states as the situation demands. For someone like you, with those wide HR swings during hockey, the interesting question isn't whether you can regulate — clearly you can. It's whether your **switching speed is getting better or worse** over time, and whether you can see that in the data. You already had that insight when you said lowering heart rate isn't the goal. This paper gives the framework for why that instinct is right.

VALIDATION · RCT, N=547

## Breathing Outperforms Physical Counterpressure

Mennitto et al. · Transfusion · 2018 · RCT

A nice large RCT that directly compared respiratory control versus applied muscle tension versus a combination. The breathing group was **significantly less likely to require treatment** for vasovagal symptoms, and the effect was actually stronger in people with lower baseline anxiety — which, given your profile, is worth noting. The way I think about it: counterpressure maneuvers are useful in the moment, during a prodrome. But breathing training changes the baseline over time, so episodes become less frequent in the first place. Both have a role, but they're solving different problems.

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### A COUPLE OF THINGS WORTH THINKING ABOUT

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You raised some good questions in your conversation with Jason, and I wanted to share how I think about them — not as a pitch, just as context from someone who sits in this world every day.

## It Depends on What You're Trying to Solve

Beta blockers are great tools — I use them clinically. But they work by blunting the sympathetic response, and for someone whose episodes are driven by a **parasympathetic surge** (which is what vasovagal syncope is), the picture gets more complicated. Your resting heart rate of ~65 with vasovagal episodes suggests you probably have fairly high baseline vagal tone. That's actually a good thing from a cardiovascular standpoint — but it means when the vagal brake fires at the wrong moment, the drop can be dramatic. A beta blocker would lower your HR ceiling during hockey without really addressing the **baroreflex** — the feedback loop that detects a blood pressure drop and corrects it before you feel the prodrome. What's interesting about the Bellocchi study is that baroreflex sensitivity specifically is what improved with breathing training. So the question isn't really beta blockers versus breathing — it's whether you want to dampen the response or train the system to respond more accurately. Both have a place. They're just doing different things.

## Your Experience Is More Common Than Most People Realize

You went through an extensive neuro and inner-ear workup and came out without a clear diagnosis. That experience is, unfortunately, incredibly common. Autonomic dysfunction tends to hide behind other labels — vertigo, anxiety, fatigue, exercise intolerance, unexplained syncope. The estimates in the literature are striking: roughly **70 million Americans** have some form of dysautonomia, POTS alone affects 1-3 million, and vasovagal syncope is the single most common cause of fainting, touching **up to 40% of people** at some point in their lives. We chose to start with nurse anesthetists and anesthesia providers because they already understand this physiology deeply — they manage other people's autonomic nervous systems every day. They're a natural starting point for building clinical credibility. But the tools we're developing — for measuring and training autonomic flexibility — are relevant to a much wider group of people who are going through exactly what you went through: something is off, the standard workup doesn't quite explain it, and the answers end up being closer to physiology than pharmacology.

### SPECIFIC TECHNIQUES TO EXPLORE

#### 1 Resonance Frequency Breathing

~5.5–6 breaths per minute. This maximizes respiratory sinus arrhythmia and entrains the baroreflex. A 2025 meta-analysis found that on-screen biofeedback with shorter daily sessions (<20 min) produced the strongest effects. The key is finding your individual resonance frequency — it varies person to person, and it's the frequency where your heart rate variability is maximally amplified.

#### 2 Inspiratory Muscle Training

Resistance breathing using a threshold device. Directly strengthens the diaphragm, improves the mechanical pump affecting venous return, and — per the Bellocchi data — improves baroreflex sensitivity. This addresses both the Valsalva avoidance you're already practicing and the underlying autonomic responsiveness.

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## Salt & Hydration Optimization

Current guidelines for autonomic dysfunction: 10–12g salt/day with 2–3L fluid to expand plasma volume. For my kid, this was foundational — it gave the nervous system enough hemodynamic buffer to stop overcorrecting. Simple. Boring. Effective.

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## Graduated Exercise Reconditioning

The Modified Dallas Protocol starts with recumbent exercise and progressively loads the autonomic system. A 2025 systematic review found HR improvements across all studies, with 86% of adolescents reporting resolved or significantly improved symptoms at 5-year follow-up. The principle scales — structured cardiovascular load teaches the baroreflex to respond faster. You're already doing this with hockey. The question is whether a structured ramp alongside it accelerates the improvement.

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WHAT WE'RE BUILDING

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## 60 seconds. Your heartbeat. 21 science-backed breathing techniques.

You asked what the product does. Pausality is a breathing app that uses real-time heart rate biofeedback — through your AirPods or Apple Watch — during 60-second guided sessions. 21 sessions, 21 days, each one building on the last. Layered breathwork, sensory cues, and biometric feedback so you can actually see your nervous system responding in real time. No guesswork.

I think of it as learning to play your cranial nerves like a violin. You start with open strings — simple breathing patterns, getting familiar with what your body is doing. Over 21 days, you develop a whole spectrum of techniques for working with your autonomic nervous system with increasing precision. By the end, you're making fine adjustments in real time based on what your body is actually telling you.

We built it for nurse anesthetists first. The early feedback from clinicians has been really encouraging — here's a note from one of our beta testers:

*"Pediatric anesthesia can be incredibly high-stress — we're trained to stay focused and composed, but sometimes you don't realize how much tension you're holding until you release it. While waiting for the surgeon to finish a pre-op for a pediatric dental case, I took one minute to use the Pausality app. That short pause completely reset my mindset. I felt mentally clear, steady, and ready for the six cases that followed. It's a simple tool, but in our line of work, that kind of reset can make all the difference."*

— CRNA, Pediatric Dental Anesthesia Provider

I'm happy to send the full citations with DOI links for any of the studies above — and honestly, I'm just as happy to talk through any of this over a call. The physiology, the product,

the experience with my kid. I find these conversations genuinely valuable. When someone is paying attention to their own body the way you are, there's a lot to learn from each other.

Thanks for sharing your experience with Jason. It meant something to hear it.

## Randy Graybeal, MSN, CRNA

Founder & CEO

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[From relaxation to regulation](#)

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All citations from PubMed · Full DOI links available on request